DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 9

# Cloud expert

Profession 9 of 10 in Development and IT.

## What the profession means

A cloud expert sets up and runs computing, storage, databases and AI services on cloud platforms such as AWS, Microsoft Azure and Google Cloud. The work covers accounts and access, hosting, networks, cost control, security and monitoring.

## What professionals do every day

Designs the architecture, provisions services, sets permissions with least privilege, deploys applications, monitors costs and performance, and plans backups and recovery.

## Why businesses hire them, and what problems they solve

Applications need reliable hosting, secure access and costs that stay predictable. A cloud expert makes the platform run well and shows where the money goes.

- Servers that go down or cost more than planned.
- Access that is too broad and a security risk.
- Data that is not backed up or cannot be restored.

## Skills you need

- Cloud basics: regions, compute, storage, databases, networks.
- Identity and access management (least privilege).
- Infrastructure basics and deployment.
- Cost management and budgets.
- Monitoring, backups and disaster recovery.

## Tools (free and paid)

| Tool | What it is for | Free option | Paid option | Official link (not checked live) |
|---|---|---|---|---|
| AWS | Broad cloud platform | Free tier for many services | Pay per use | aws.amazon.com/free |
| Microsoft Azure | Cloud and AI services | Free account credits | Pay per use | azure.microsoft.com/free |
| Google Cloud | Cloud and AI services | Free tier | Pay per use | cloud.google.com/free |
| Vercel | Hosting for web apps | Hobby tier | Paid plans | vercel.com |
| Netlify | Hosting for static sites and functions | Free tier | Paid plans | www.netlify.com |

## Learning sequence: beginner, intermediate, advanced

| Level | What you can do when you finish |
|---|---|
| **Beginner** | Host a static site on a free tier with HTTPS. |
| **Intermediate** | Set a budget alert and a spending limit. |
| **Advanced** | Create a database with backups and test one restore. |

## Projects by level

- **Beginner:** Deployment pipeline for a web app.
- **Intermediate:** Cost report and optimisation for a client.
- **Advanced:** Backup and recovery plan with a tested restore.

## Client-ready portfolio project

A hosted application for a fictional business with least-privilege access, a budget alert, automated backups, one tested restore and a one-page cost summary.

## Worked example

**Worked example.** A small app needs a database and a web server. The cloud expert gives the app its own role with permission only for that database, sets a budget alert at a fixed amount, and schedules a backup. The restore is tested, not assumed.

## Assignment with full solution

ASSIGNMENT

### Budget alert and a tested restore

**Business problem:** A client's app runs on a cloud account. The client fears unexpected bills and data loss.

**Steps**

1. Create a budget with an alert at 50% and 80% of the monthly limit (set it in the provider's billing console).
2. Create a role for the application with access to one database only.
3. Turn on automated backups and record the schedule.
4. Restore a backup to a test copy and check three records match.
5. Write a one-page summary: limits, owners, backup schedule and the restore steps.

**Expected output:** Alerts configured, a restore test log with matching records, and a one-page summary.

SOLUTION (ANSWER KEY)

Alerts and least-privilege roles limit both cost and damage. A backup you have never restored is a hope, not a plan. Exact console steps change; follow the provider's current documentation.

## Common mistakes

- Using the root or admin account for everyday tasks.
- Leaving test resources running and paying for them.
- Storing backups in the same place as the live data.
- Never testing a restore.

## Freelancing opportunities

- Hosting and deployment for small businesses and startups.
- Cost reviews and cloud clean-ups.
- Freelance briefs for 'AWS', 'Azure', 'Google Cloud', 'deployment'.

## Learning resources

**LEARNING RESOURCES (VERIFY BEFORE USE)**

- AWS Training (official)
- Microsoft Learn: Azure
- Google Cloud Skills (official)

Part of **Master Freelancing: AI and Automation**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.